Goal: Interpret the definite integral as the (signed) area underneath the graph of a function.
Recall Left/Right Riemann sums (L_N and R_N) are used to approximate the (signed) area underneath the graph of a function.

Question: What happens when we take $N \to \infty$? See what happens with a Desmos presentation (link here).

$$\lim_{N\to\infty} L_N = \lim_{N\to\infty} R_n = \text{The exact area}$$

Definition. The **(definite) integral** of a function f from a to b is denoted as follows:

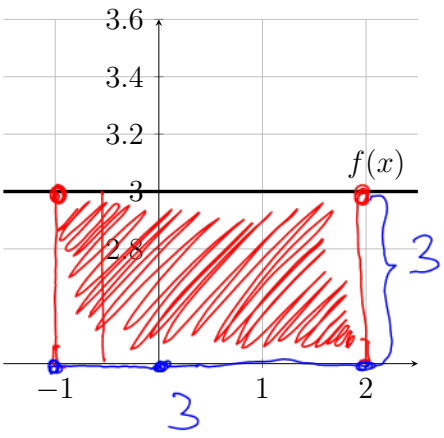
$$\int_a^b f(x)\,dx \overset{\text{def}}{=} \lim_{N\to\infty} L_N = \lim_{N\to\infty} R_n$$

(b: Upper bound; a: Lower Bound)

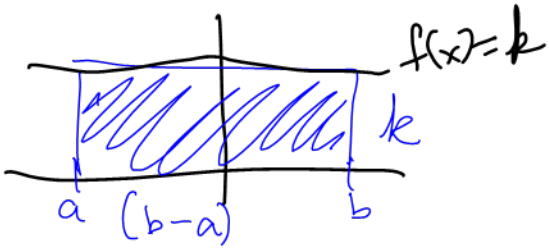
$$= \text{The signed area between the graph of } f \text{ and the } x\text{-axis on } [a,b]$$

a and b are called the upper and lower limits (or bounds) of integration, respectively. If the integral is finite, we say f is (Riemann) integrable on $[a, b]$.

Example 1. Compute $\int_{-1}^{2} 3\,dx$ ("The integral of 3 from -1 to 2")



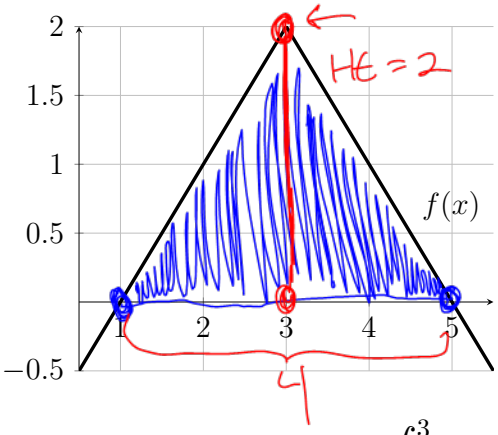
$$\int_{-1}^{2} 3\,dx = \text{Area of the Rectangle}$$
$$= [\text{Base}]\,[\text{Height}]$$
$$= [3][3]$$
$$= 9$$



NOTE: In general, if k is a constant:

$$\int_a^b k\,dx = k(b-a)$$

Example 2. Compute $\int_1^5 (2-|x-3|)\,dx$



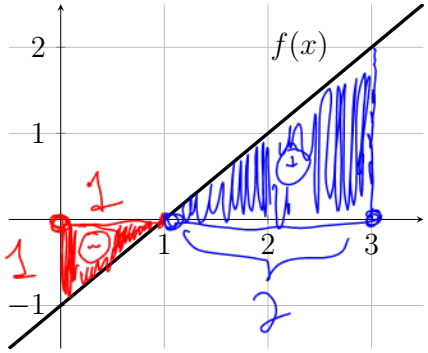
$\int_1^5 (2-|x-3|)\,dx =$ Area of the Triangle

$= \frac{1}{2}[\text{Base}][\text{Height}]$

$= \frac{1}{2}[4][2]$

$= 4$

Example 3. Compute $\int_0^3 (x-1)\,dx$



$\int_0^3 (x-1)\,dx =$ Area of Yellow Triangle − Area of Blue Tri.

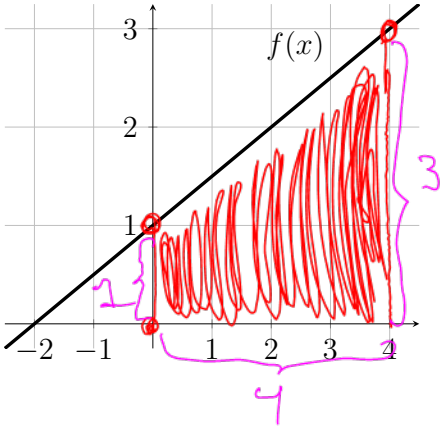
$= \frac{1}{2}(2)(2) - \frac{1}{2}(1)(1)$

$= 2 - \frac{1}{2}$

$= \frac{3}{2}$

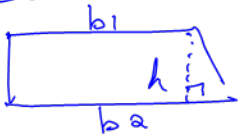
Example 4. Compute $\int_0^4 \left(\frac{1}{2}x+1\right)\,dx$

(a) METHOD 1:



Area of a trapezoid:

$A = \frac{1}{2}(b_1+b_2)\,h$

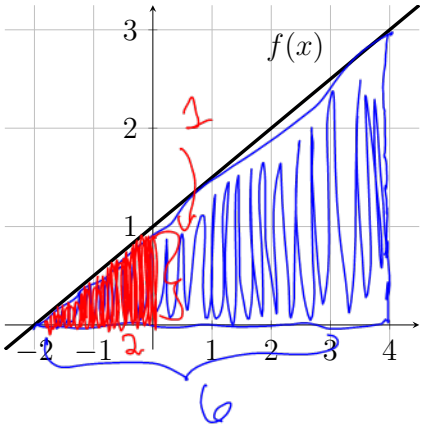


$\int_0^4 \left(\frac{1}{2}x+1\right)\,dx =$ Area of Trapezoid

$= \frac{1}{2}[1+3][4]$

$= 8$

(b) METHOD 2:



$$\int_0^4 \left(\frac{1}{2}x+1\right)\,dx =$$

Area of Yellow Triangle − Area of Blue Triangle

$$= \frac{1}{2}(6)(3) - \frac{1}{2}(2)(1)$$

$$= 9 - 1$$

$$= 8$$

NOTE (Additivity of Integrals Example):

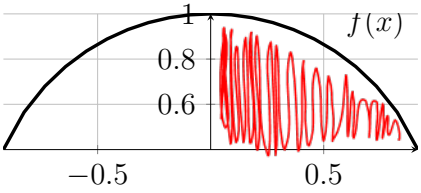
Rewriting METHOD 2 using integrals, we get:

$$\int_0^4 \left(\frac{1}{2}x+1\right)\,dx = \int_{-2}^{4}\left(\frac{1}{2}x+1\right)dx - \int_{-2}^{0}\left(\frac{1}{2}x+1\right)dx$$

I.e.,

$$\int_{-2}^4 \left(\frac{1}{2}x+1\right)\,dx = \int_{-2}^{0}\left(\frac{1}{2}x+1\right)dx + \int_{0}^{4}\left(\frac{1}{2}x+1\right)dx$$

Example 5. Compute $\int_0^1 \sqrt{1-x^2}\,dx$



R^2

$y = \sqrt{1-x^2}$

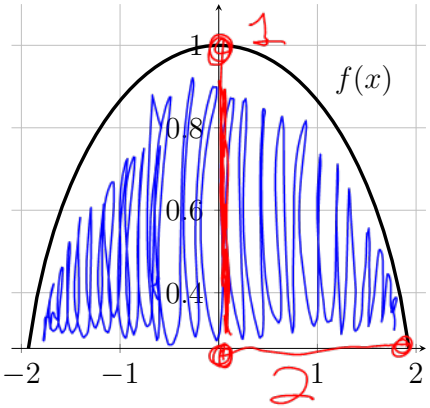
$y^2 = 1 - x^2$

$x^2 + y^2 = 1$

$$\int_0^1 \sqrt{1-x^2}\,dx = \frac{1}{4}\left[\text{Area of Circle w/ Rad. 1}\right]$$

$$= \frac{1}{4}\cdot\pi(1)^2 = \frac{\pi}{4}$$

Example 6. Compute $\int_{-2}^{2} \sqrt{1-\frac{x^2}{2^2}}\,dx$



$$y = \sqrt{1-\frac{x^2}{2^2}}$$

$$y^2 = 1 - \frac{x^2}{2^2}$$

$$\frac{x^2}{2^2} + \frac{y^2}{1^2} = 1$$

$$\int_{-2}^{2} \sqrt{1-\frac{x^2}{2^2}}\,dx$$

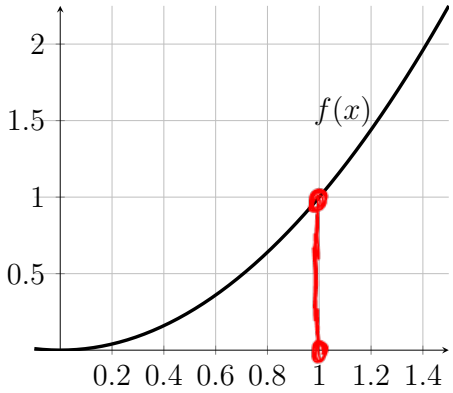
$$= \frac{1}{2}\left[\text{Area of Ellipse}\right]$$

$$= \frac{1}{2}\cdot\pi\cdot 2\cdot 1$$

$$= \pi$$

$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \Rightarrow$ Area is πab

Example 7. Compute $\int_1^1 x^2\ dx$



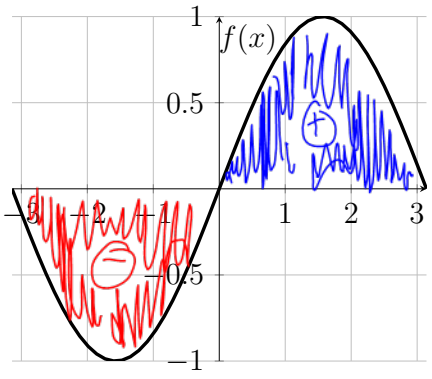
$\int_1^1 x^2\,dx$ = Area of the Line Segment

$= 0$

NOTE: In general,

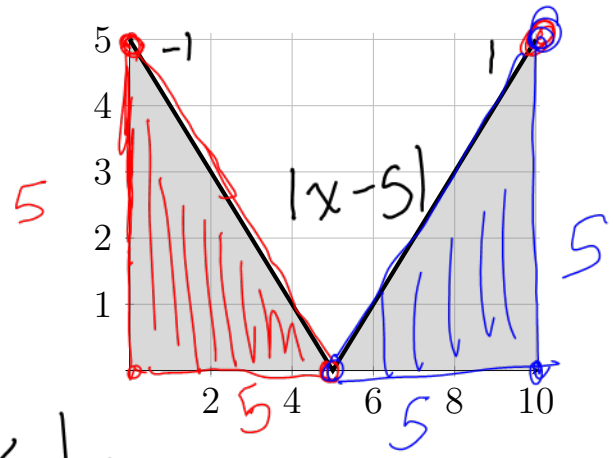
$$\int_a^a f(x) = 0$$

Example 8. Compute $\int_{-\pi}^{\pi} \sin x\ dx$



$\int_{-\pi}^{\pi} \sin x\,dx = 0$

Example 9. Consider the shaded region below:



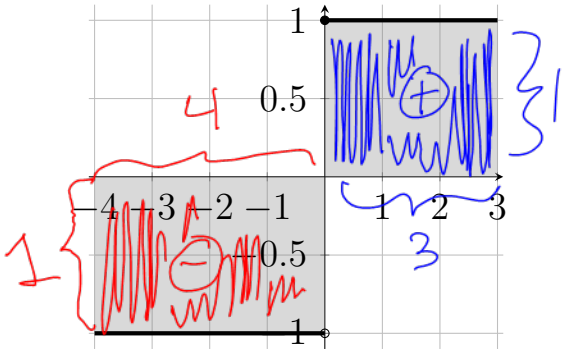
(a) Describe the area of the shaded region using an integral:

$$\int_0^{10} |x-5|\,dx$$

(b) Compute the area of the shaded region.

$$\int_0^{10} |x-5|\,dx = \frac{1}{2}(5)(5) + \frac{1}{2}(5)(5) = 25$$

Example 10. Consider the shaded region below:



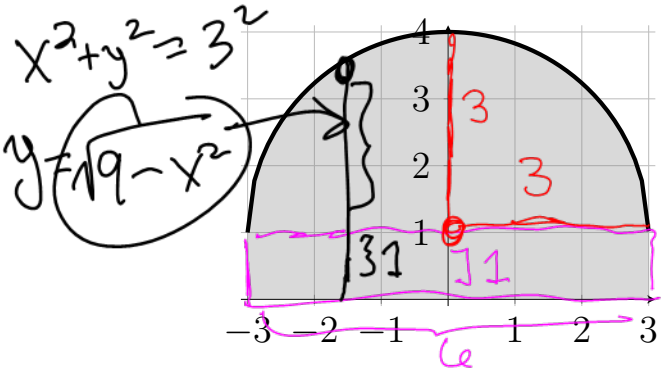
(a) Describe the area of the shaded region using an integral:

$$\int_{-4}^{3} f(x)\,dx \text{ where } f(x) = \begin{cases} -1 & x<0 \\ 1 & x\geq 0 \end{cases}$$

(b) Compute the area of the shaded region.

$$\int_{-4}^{3} f(x)\,dx = 3(1) - 4(1) = -1$$

Example 11. Consider the shaded region below:



(a) Describe the area of the shaded region using an integral:

$$\int_{-3}^{3} \left(1 + \sqrt{9-x^2}\right) dx$$

(b) Compute the area of the shaded region.

$$\int_{-3}^{3} \left(1 + \sqrt{9 - x^2}\right) dx = 6(1) + \frac{1}{2}\pi(3)^2 = 6 + \frac{9}{2}\pi$$

NOTE (Linearity of Integrals Example):

Rewriting our work using integrals, we see:

$$\int_{-3}^{3} (1 + \sqrt{9 - x^2})\, dx = \int_{-3}^{3} 1\, dx + \int_{-3}^{3} \sqrt{9 - x^2}\, dx$$